Template: SWOT Analysis

A SWOT analysis is a compilation of your company's strengths, weaknesses, opportunities and threats. The primary objective of a SWOT analysis is to help organizations develop a full awareness of all the factors involved in making a business decision.

INTERNAL

Strengths

- *What differences do your concept make?*
- *What difference do your concept make? - Why are customers willing to pay?*
- *What distinguishes your concept from competitors' concepts (why is yours preferable)?*
- *Is it difficult to copy your concept?*

Weaknesses

- *Do you miss resources or knowledge to realize your concept?*
- *Is there a risk that your concept is used in a different (and non-optimal) way from what was intended?*
- *What is your weakest point with your potential customers?*

EXTERNAL

Opportunities

Are there societal trends that could make your concept more popular?
Can future technologies make your concept even stronger?
Are there uninvestigated potential partners?

Threats

Are there societal trends that may potentially affect your concept negatively? Is it easy for competitors to damage your position?
What could impact your customers and/or users to not using your concept in the future?

INTERNAL

POSITIVE

NEGATIVE

STRENGTHS - Utilizes local solar energy and storage to allow the boiler to run on low cost electricity, enabling affordable and sustainable heating. - Relies on rented residential rooftops and distributed small PV parks, protecting valuable agricultural land and natural areas from overdevelopment. - Expands local electrical capacity to support BEOF's electrification goals. - Our plans needs multi-stakeholder trust, municipal alignment, and deep integration with BEOF's current boiler infrastructure make it challenging for outside competitors to imitate.	*WEAKNESSES* - Our plan relies heavily on the solar panels, and during winter the production will be very low. - The current timeline requires is approx. 4 years long, so stakeholders face a significant delay before seeing any return on investment. - There is a risk that the system's battery storage could eventually be prioritized for selling power to the broader market for profit, rather than its intended purpose..
- Rising global interest and investment in. green/sustainable energy - Possibility to increase collaborations with locals for Bornholm's benefit. - The system could be expanded with more epsolar panels, batteries, or smart energy management as technology improves. *OPPORTUNITIES*	- Solar energy production can be low during cloudy periods and winter, creating potential energy shortages. - Government policies, subsidies, or electricity prices could change and affect the economic viability of the system. - Refusal to rent out rooftops to the project if locals are against it. *THREATS*

EXTERNAL

Template: Implementation Plan

Key Partners and Resources

Who are needed for our concept to be realized (collaboration companies, organizations, politicians…)? Which Key Activities do these partners perform?

What are the partners motivations for engaging in a partnership? Are there any risks connected to the partnerships? (If 'yes', which actions can be made to minimize those risks?)

BEOF owns and operates the boiler and battery storage and buys the injected solar electricity. It's motivated by its biomass-reduction targets and cheaper, more stable electricity. Risk: BEOF may favor its own large-scale generation over many small rooftop contracts, mitigated by framing rooftop solar as a low-capital complement, not a competitor.

Nexø residents/building owners lease their roofs or self-fund panels, motivated by rental or electricity-sale income and community participation. Risk: low uptake if terms feel unattractive, mitigated with simple, transparent contracts (modeled on Project Sun) and community info sessions.

Bornholm Regional Municipality provides planning/permitting authority and political backing, motivated by climate targets and local economic activity. Risk: shifting political priorities, mitigated by aligning the pitch with the municipality's existingsustainability goals.

Local investors; this project should be open for both residents but also companies like Ocean Prawns to invest in. This will help boost local ties and give a sense of common ownership, along with gaining capital for the project. The risk would be, that the rooftop solar park does not make any money, but to mitigate this risk a similar deal that was made with the Baltic energy island, and the government could be sought.

Costumers and Costumer Relationships

Who are going to buy our concept? How do you establish a contact to potential customers? What type of relationship do we need to establish to our costumers?

When a customer is on board, how often do you need to be in contact with your customer? Who is responsible for maintaining the concept?

BEOF is our main customer: it buys the solar electricity from the rented rooftops to run its hybrid boiler system. Contact isdirect, since BEOF is already our project partner. The relationship is a long-term contractual one, since it involves ongoing electricity purchases rather than a one-time sale.

Nexø residents/building owners are a second customer group: they join by leasing their roof or selffunding panels, in exchange for rental or electricity income. Contact happens through community outreach: info sessions, homeowner associations, and local channels. This relationship needs to be simple and trust-based, with clear contracts and responsive support.

Once onboard, BEOF needs regular, ongoing contact as the system scales, while residents need lighter checkins and quick support when issues come up. A small dedicated team, either within BEOF or a partnering solar operator, should be responsible for maintaining the concept, since managing rooftop contracts and residents isn't BEOF's core business.

How do you sell your concept

How do our costumers get aware of our concept? Which concerns/critical questions may the potential costumers have regarding buying your concept? How big is the market?

Awareness: Residents learn about the concept through local channels that already reach the Nexø community: municipality announcements, local newspapers (like Bornholm.nu/Tidende), community meetings, and direct outreach via homeowner associations. BEOF, as an already-engaged partner, is reached directly through the project itself rather than needing marketing.

Concerns customers may have:
Residents: Will renting my roof damage it or affect my home insurance? What happens if I sell my house during the contract? Is the rental income actually worth the hassle? How long is the contract?
BEOF: Is the electricity supply reliable enough to depend on? Will managing many small rooftop contracts be more effort than it's worth compared to one large solar installation? Is there enough grid capacity to actually use this energy?

Market size: The market is bounded by the number of suitable rooftops in Nexø with enough south-facing, unshaded roof area to host panels, realistically a subset of the town's residential and commercial buildings. This is a small, local, and finite market rather than a scalable mass-market product, but that matches the concept's purpose: solving a specific local grid-capacity problem in Nexø, not building a nationwide solar business.

Template: Expanded Mental Model Evaluation

DESIRABILITY

Will this solution fill a need? Will it fit into people's lives? Will it appeal to them? Will they actually want it?

Residents can benefit from cheaper and more green district heating. by utilizing low-cost solar electricity and storing energy for later use. Furthermore, the solution utilizes unused rooftop area instead of occupying additional land in Bornholm, helping to preserve the local landscape and nature. This is particularly relevant, as preserving Bornholm's nature and landscape was highlighted as important across our resident interviews. These economic and environmental benefits could increase residents' acceptance and willingness to participate in the solution.

FEASIBILITY

Is the technology needed to power the design solution available or within reach? How long will this take? Can the organization actually make it happen?

The solution is technically feasible because the required technology for the project, including rooftop solar panels, batteries, boilers and heat storage, are already available and proven. The main challenges are identifying suitable rooftops, establishing agreements with building owners, obtaining municipal approvals and integrating the solution with the existing energy infrastructure. Based on our implementation plan, the process is expected to take several years. A smaller pilot project could therefore be used to test the solution before implementing it on a larger scale.

VIABILITY

Will the design solution align with the client's business? How is the economic situation: Does this solution honor the client's budget? What will the return on the investment look like?

The solution could make economic sense because existing rooftops can be used instead of using more land for solar parks. This could reduce some costs while also creating more space for renewable energy production. However, there are still costs related to installing the solar panels, renting rooftops, batteries and maintenance. Therefore, more calculations are needed to see if the solution will be profitable in the long term. The business case will depend on whether the value created from the solar electricity and heat is greater than the cost of installing, operating and maintaining the system. But overall, it looks like the project has a great potential for now.

Template: Expanded Mental Model Evaluation

RESPONSIBILITY

Are stakeholders satisfied with the solution or would any of them complain?
If my product or service is hugely successful, what would that look like? Which positive or negative consequences might that have?
Imagine it is ten years in the future, and your product or service is in the headlines as being destructive for society. What happened?

The majority of the stakeholders would be satisfied with the solution, as it focusses an accommodating the needs of both BEOF, residents and local businesses. Some of the residents may choose not to be a part of the project, and may be annoyed by the construction, but luckily the construction part for putting up solar panels is very short.
BEOF would have their expanded energy grid, and local business would have the opportunity to expand or even invest in the project.
The municipality could benefit from progress towards local climate goals, while using rooftops instead of open land helps preserve Bornholm's nature. However, they would face an expense considering the project.

If the project is hugely successful, the residents of Nexø all participate in the project, the project is then implemented with both heat storage and batteries supplying the area with sufficient heat and electricity, and the Rooftop Solar Park concept is spread to other small villages on the island until Bornholm is self sufficient in both heating and electric energy.

If our project headlines for being destructive, then the economy of the project collapsed, while having all the residents and local businesses as investors. This would not only be a major blow to the concept but also to the economy of Bornholm which already sports one of the lowest person incomes in the country.

SUSTAINABILITY

Is my creation aligned with principles used by nature in the sustenance of life on earth?
What more could you improve in terms of sustainability if you had no resource limitations?
Would I be proud to tell my grandchildren that I was part of bringing this product or service into the world?

The solution supports sustainability by increasing the use of renewable solar energy while making use of existing rooftop space. This reduces the need to use additional land for solar parks and helps preserve Bornholm's nature and landscape. Batteries and heat storage can also help make better use of the produced solar energy by storing excess energy for later use. However, producing solar panels and batteries also requires materials and resources and of course these components will eventually need to be replaced or recycled. Therefore, in the future, the solution can be improved by choosing the components well, meaning choosing components with a long lifetime, as sustainable as possible.

Template:

Expanded Mental Model

